

OXVERSE ACADEMY

Mobile App Development Curriculum

Professional Diploma · Cohort-based · Project-driven

DURATION

3 Months (12 Weeks)

TOTAL WEEKS

12

LEVEL

Beginner to Industry Ready

PROJECTS

12+

CAPSTONE

1 Published Cross-Platform App

PROGRAMME GOAL

Train learners to design, build, test, and publish production-grade cross-platform mobile applications using React Native, Expo, and TypeScript, covering native device APIs, backend integration, payments, animations, and real app store deployment.

OVERVIEW

A 12-week intensive mobile development diploma taking learners from JavaScript/TypeScript fundamentals through React Native and Expo architecture, navigation, state management, native device capabilities, offline data, authentication, backend and payment integration, advanced animations, testing, performance tuning, and finally publishing a real app to the Google Play Store and Apple App Store as a graded capstone.

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Course Outline

1. **Week 1:** TypeScript & Mobile Development Foundations
2. **Week 2:** React Native Core Components & Styling
3. **Week 3:** Navigation & App Architecture
4. **Week 4:** State Management & Data Fetching
5. **Week 5:** Native Device APIs
6. **Week 6:** Offline Storage & Data Persistence
7. **Week 7:** Authentication & Backend Integration
8. **Week 8:** Payments, Monetization & Third-Party Services
9. **Week 9:** Animations, Gestures & Advanced UI
10. **Week 10:** Testing, Debugging & Performance Optimization
11. **Week 11:** Native Modules, CI/CD & App Store Preparation
12. **Week 12:** Capstone: Build & Publish a Cross-Platform App

TypeScript & Mobile Development Foundations

Establish strong JavaScript/TypeScript fundamentals and set up the modern React Native + Expo development environment.

LEARNING OBJECTIVES

- Write type-safe TypeScript code using interfaces, types, and generics
- Set up a full React Native/Expo development environment on macOS/Windows/Linux
- Understand the mobile app architecture and how JS bridges to native code
- Run and debug apps on iOS simulator, Android emulator, and physical devices

DETAILED TOPIC BREAKDOWN

1.1 TypeScript Fundamentals for Mobile

Primitive types, arrays, tuples, and enums Interfaces vs type aliases Union and intersection types

Optional and readonly properties Generics in functions and interfaces Type narrowing and type guards

Utility types: Partial, Pick, Omit, Record Strict mode and tsconfig.json configuration Async/await typing and Promise generics

Modules, imports, and path aliases

1.2 React Native & Expo Ecosystem

History and architecture of React Native (Bridge vs New Architecture/Fabric) What Expo is and Expo Managed vs Bare workflow

Expo SDK overview and versioning Expo CLI vs React Native CLI EAS (Expo Application Services) overview

JavaScript thread vs UI thread vs native modules Metro bundler basics

Comparing React Native to Flutter and native development

1.3 Development Environment Setup

Installing Node.js, Watchman, and package managers (npm/yarn/bun) Installing Xcode and iOS Simulator (macOS)

Installing Android Studio, SDKs, and AVD emulator Creating a new Expo project with create-expo-app

Configuring app.json/app.config.ts Running on physical devices with Expo Go

Setting up VS Code with ESLint, Prettier, and React Native extensions Git repository setup and .gitignore for RN/Expo projects

1.4 Debugging & Dev Tools

React Native Debugger and Flipper basics Console logging strategies and remote debugging

Fast Refresh and hot reloading behavior Inspecting element trees with Dev Menu

Common setup errors and troubleshooting (SDK mismatches, pod install issues) Using expo doctor to diagnose project health

HANDS-ON EXERCISES

- Convert 10 plain JavaScript functions into strictly typed TypeScript
- Build typed interfaces for a User, Product, and Order model
- Initialize 3 separate Expo projects with different templates
- Run the same app on iOS simulator, Android emulator, and Expo Go on a physical phone

ASSIGNMENTS

- Set up a fully working local dev environment and submit screenshots of the app running on all three targets
- Write a TypeScript utility library (10+ functions) with full type coverage and no 'any' usage

PROJECTS

- Scaffold and configure a 'Hello Mobile World' Expo TypeScript starter app with custom app icon, splash screen, and typed constants file

WEEKLY LEARNING OUTCOMES

- Comfortable writing idiomatic TypeScript
- Fully configured cross-platform mobile dev environment
- Understanding of RN/Expo architecture and tooling
- Ability to debug and troubleshoot setup issues independently

WEEKLY ASSESSMENT

Environment setup review + TypeScript coding test (20 typed exercises, timed)

React Native Core Components & Styling

Master the fundamental building blocks of React Native UIs and responsive styling systems.

LEARNING OBJECTIVES

- Build screens using core React Native components
- Apply Flexbox layout for responsive UI across screen sizes
- Style components using StyleSheet and design tokens
- Handle user input and touch interactions

DETAILED TOPIC BREAKDOWN

2.1 Core Components

View, Text, Image, ScrollView, SafeAreaView TextInput and controlled inputs TouchableOpacity, Pressable, and touch feedback

FlatList and SectionList for performant lists Modal and Alert components ActivityIndicator and loading states

KeyboardAvoidingView and keyboard handling Platform-specific components with Platform.OS

2.2 Flexbox & Layout System

Flex direction, justify-content, align-items in RN's flexbox Absolute vs relative positioning Dimensions API and responsive units

useWindowDimensions hook for orientation changes Percentage-based vs fixed layouts

Building responsive grids with FlatList numColumns SafeAreaContext and notch/status bar handling

Aspect ratio and image scaling

2.3 Styling Systems

StyleSheet.create and style objects Style composition and arrays Theming with color palettes and design tokens

Dark mode with useColorScheme Custom fonts with expo-font NativeWind (Tailwind for React Native) setup and usage

Shadow and elevation cross-platform styling Reusable styled component patterns

2.4 Forms & User Input

Controlled TextInput with validation Building a login form UI Handling keyboard types and input masking

Form libraries: React Hook Form for React Native Displaying validation errors inline Picker and Switch components

Date/time pickers with @react-native-community/datetimetypepicker

HANDS-ON EXERCISES

- Recreate 5 real app UI screens (Instagram profile, WhatsApp chat list, food delivery card) pixel-for-pixel
- Build a responsive grid layout that adapts to phone and tablet
- Implement dark mode toggle across an entire screen
- Build a multi-field signup form with client-side validation

ASSIGNMENTS

- Design and implement a reusable design system (Button, Card, Input, Badge components) with TypeScript props
- Build a fully responsive product listing screen using FlatList and custom styling

PROJECTS

- 'Social Feed UI Clone' - a static but fully responsive and styled social media feed screen with posts, likes, and comments UI

WEEKLY LEARNING OUTCOMES

- Fluency with all core React Native components
- Strong command of Flexbox layout for mobile
- A personal reusable component/design system library
- Ability to translate Figma designs into pixel-accurate RN UI

WEEKLY ASSESSMENT

UI clone challenge graded against a provided design mockup

Navigation & App Architecture

Implement multi-screen navigation patterns using React Navigation and structure scalable app architecture.

LEARNING OBJECTIVES

- Implement stack, tab, and drawer navigation
- Pass and type-check navigation parameters
- Structure a scalable folder architecture for mobile apps
- Handle deep linking and universal links

DETAILED TOPIC BREAKDOWN

3.1 React Navigation Fundamentals

Installing and configuring React Navigation with Expo Native Stack Navigator setup and screen options
Bottom Tab Navigator with custom icons Drawer Navigator for side menus Nesting navigators (stack inside tabs)
Header customization and hiding headers Screen transitions and animations config Navigation lifecycle events (focus, blur)

3.2 Typed Navigation & Params

Defining ParamList types for stacks and tabs Typing useNavigation and useRoute hooks
Passing and validating route parameters Navigating programmatically (navigate, push, replace, goBack)
Resetting navigation stack on logout Modal screens and presentation styles Nested navigator type composition

3.3 App Architecture & Project Structure

Feature-based folder structure vs type-based structure Separating screens, components, hooks, services, and utils
Environment variables with expo-constants and .env files App-wide constants and theming configuration
Absolute imports and path aliases (tsconfig paths) Creating a shared api/ service layer
Error boundary components for crash isolation

3.4 Deep Linking & App Flow

Configuring URL schemes with app.json Handling deep links with expo-linking
Universal links for iOS and App Links for Android basics Conditional navigation flows (onboarding vs authenticated vs guest)
Splash screen control with expo-splash-screen Persisting and restoring navigation state

HANDS-ON EXERCISES

- Build a 4-tab app with nested stack navigators in each tab
- Implement typed navigation between 6 screens with parameter passing
- Create an onboarding flow that only shows on first app launch
- Add a working deep link that opens a specific product detail screen

ASSIGNMENTS

- Architect a full app skeleton with auth flow, tab navigation, and modal screens, fully typed with TypeScript
- Document your app's folder architecture and navigation map as a diagram

PROJECTS

- 'Multi-Screen Marketplace App Shell' - full navigation skeleton with onboarding, auth stack, tab navigation (Home, Search, Cart, Profile), and deep linking to product pages

WEEKLY LEARNING OUTCOMES

- Confident implementation of complex navigation trees
- Type-safe navigation across the entire app
- A scalable, maintainable project architecture
- Working deep linking implementation

WEEKLY ASSESSMENT

Architecture review + live navigation flow demo

State Management & Data Fetching

Manage local and global app state and fetch/cache remote data efficiently.

LEARNING OBJECTIVES

- Manage complex local state with hooks
- Implement global state management with Zustand/Redux Toolkit
- Fetch, cache, and synchronize server data with React Query
- Handle loading, error, and empty states gracefully

DETAILED TOPIC BREAKDOWN

4.1 React State & Hooks Deep Dive

useState, useEffect, useReducer patterns useMemo and useCallback for performance Custom hooks for reusable logic

Context API for shared state (theme, auth) Avoiding prop drilling and re-render pitfalls

useRef for mutable values and imperative handles

4.2 Global State Management

Zustand store setup and slices Redux Toolkit setup: store, slices, reducers Async thunks for side effects

Selecting and memoizing state selectors Persisting state with AsyncStorage middleware

Choosing Zustand vs Redux vs Context for a project DevTools integration for state debugging

4.3 Data Fetching with React Query

Setting up TanStack Query in a React Native app useQuery for GET requests and caching

useMutation for POST/PUT/DELETE operations Query invalidation and refetching strategies Optimistic updates

Pagination and infinite queries with useInfiniteQuery Handling loading, error, and stale data states

Background refetching and network status awareness

4.4 API Integration Patterns

Structuring an Axios/fetch API client layer Interceptors for auth tokens and error handling Typing API responses with TypeScript

Environment-based API base URLs (dev/staging/prod) Handling network errors and retries

Mocking APIs during development with MSW or json-server

HANDS-ON EXERCISES

- Build a Zustand store managing cart state across multiple screens
- Fetch a public REST API's data with React Query and render a paginated list
- Implement optimistic UI update for a 'like' button
- Add pull-to-refresh and infinite scroll to a FlatList

ASSIGNMENTS

- Build a complete product catalog feature: fetch, cache, filter, and paginate data from a real API
- Refactor a Context-based state solution into Zustand and justify the tradeoffs in a short writeup

WEEKLY LEARNING OUTCOMES

- Mastery of local and global state management strategies
- Efficient data fetching and caching with React Query
- Robust handling of loading/error/empty UI states
- A working persisted shopping cart feature

WEEKLY ASSESSMENT

Code review of state architecture + live debugging challenge

PROJECTS

- 'E-Commerce Catalog & Cart' - product listing with infinite scroll, detail screen, cart managed globally, and persisted across app restarts

Native Device APIs

Integrate native hardware and OS-level capabilities including camera, location, sensors, and permissions.

LEARNING OBJECTIVES

- Access camera and media library from the app
- Retrieve and use device location data
- Request and manage runtime permissions properly
- Use device sensors and haptic feedback

DETAILED TOPIC BREAKDOWN

5.1 Permissions Management

- Expo Permissions API overview
- Requesting camera, location, and notification permissions
- Handling denied and blocked permission states
- iOS Info.plist usage descriptions
- Android manifest permissions configuration
- Graceful permission re-request UX patterns

5.2 Camera & Media

- expo-camera setup and live preview
- Capturing photos and switching front/back camera
- expo-image-picker for gallery selection
- Image compression and resizing before upload
- expo-av for video/audio playback and recording
- expo-media-library for saving to device gallery
- Building a QR/barcode scanner with expo-camera

5.3 Location & Maps

- expo-location for foreground and background location
- Getting current position and watching position changes
- Reverse geocoding addresses from coordinates
- react-native-maps setup and markers
- Drawing routes and polylines on a map
- Geofencing basics
- Handling location permission edge cases (denied, low accuracy)

5.4 Sensors, Haptics & Notifications

- expo-sensors: accelerometer, gyroscope, magnetometer
- expo-haptics for tactile feedback
- expo-notifications: local notifications setup
- Push notification tokens and registration
- Scheduling and canceling notifications
- Handling notification taps and deep linking from notifications
- Background tasks with expo-background-fetch

HANDS-ON EXERCISES

- Build a working QR code scanner that parses and displays scanned data
- Create a photo capture and gallery-save feature with compression
- Display the user's live location on a map with a custom marker
- Schedule a local notification that fires 10 seconds after a button press

ASSIGNMENTS

- Build a 'Check-In' feature combining camera capture, location tagging, and a map preview
- Implement a robust permissions flow with clear denied/blocked state UX for at least 3 permission types

PROJECTS

- 'Field Report App' - capture photo evidence, tag GPS location on a map, add notes, and receive a local notification confirming submission

WEEKLY LEARNING OUTCOMES

- Confident integration of camera, location, and sensor APIs
- Proper handling of runtime permissions across iOS and Android
- Working local push notification system
- A complete native-features-driven feature project

WEEKLY ASSESSMENT

Live demo of Field Report App on a physical device

Offline Storage & Data Persistence

Persist data locally, build offline-first experiences, and sync with remote sources.

LEARNING OBJECTIVES

- Store structured data locally using SQLite and AsyncStorage
- Build offline-first UX with sync queues
- Cache images and assets for offline use
- Secure sensitive data on-device

DETAILED TOPIC BREAKDOWN

6.1 AsyncStorage & Key-Value Storage

- AsyncStorage API basics (get, set, remove, merge)
- Storing and retrieving JSON objects
- MMKV as a faster alternative to AsyncStorage
- Building a typed storage wrapper/service
- Persisting user preferences and app settings
- Storage size limits and cleanup strategies

6.2 SQLite & Structured Local Data

- expo-sqlite setup and database creation
- Creating tables and running CRUD SQL queries
- Drizzle ORM or WatermelonDB overview for RN
- Migrations and schema versioning
- Querying and filtering local data
- Indexing for query performance

6.3 Offline-First Architecture

- Detecting network status with NetInfo
- Queueing mutations while offline
- Syncing local changes to server on reconnect
- Conflict resolution strategies (last-write-wins, merge)
- Optimistic local writes with background sync
- Caching API responses for offline viewing

6.4 Secure Storage & Asset Caching

- expo-secure-store for sensitive tokens/credentials
- Encrypting sensitive local data
- expo-file-system for file read/write operations
- Caching remote images with expo-image
- Managing app cache size and clearing cache
- Best practices: what never to store on-device

HANDS-ON EXERCISES

- Build a local SQLite-backed notes app with full CRUD
- Implement an offline network banner that queues actions when offline
- Store an auth token securely with expo-secure-store
- Cache a list of images for offline viewing

ASSIGNMENTS

- Convert the earlier E-Commerce Catalog project to work fully offline with cached product data and queued cart syncing
- Write a short technical doc comparing AsyncStorage, MMKV, and SQLite tradeoffs for different use cases

PROJECTS

- 'Offline Task Manager' - a fully offline-capable to-do/notes app with SQLite persistence, sync queue simulation, and secure storage for a mock auth token

WEEKLY LEARNING OUTCOMES

- Ability to design offline-first mobile experiences
- Comfort with SQLite and key-value local storage
- Secure handling of sensitive on-device data
- Working sync-queue pattern implementation

WEEKLY ASSESSMENT

Offline mode demo: disable network mid-demo and show app still functions

Authentication & Backend Integration

Implement production-grade authentication flows and connect the app to a real backend service.

LEARNING OBJECTIVES

- Implement email/password and social authentication
- Manage auth tokens, refresh flows, and protected routes
- Integrate a Backend-as-a-Service (Supabase/Firebase)
- Build and consume a REST/GraphQL API from the mobile app

DETAILED TOPIC BREAKDOWN

7.1 Authentication Fundamentals

- JWT structure and access/refresh token flow
- Secure token storage with expo-secure-store
- Building login, signup, and forgot-password screens
- Auth Context/store and protected navigation guards
- Session persistence across app restarts
- Handling token expiry and silent refresh
- Biometric authentication with expo-local-authentication

7.2 Social & OAuth Login

- Google Sign-In integration with expo-auth-session
- Apple Sign-In (required for App Store apps with social login)
- OAuth 2.0/PKCE flow fundamentals
- Handling deep-link redirects during OAuth
- Linking multiple auth providers to one account

7.3 Backend-as-a-Service Integration

- Supabase project setup: Auth, Database, Storage
- Firebase project setup: Auth, Firestore, Storage
- Row-level security policies in Supabase
- Realtime subscriptions for live data updates
- File uploads to cloud storage buckets
- Choosing between Supabase, Firebase, and a custom backend

7.4 Custom API Integration

- Consuming a Node.js/Express REST API from React Native
- GraphQL basics with Apollo Client in React Native
- Handling CORS and mobile-specific network configs
- Environment configuration for dev/staging/prod backends
- Error handling and standardized API error responses
- Rate limiting and request retry strategies

HANDS-ON EXERCISES

- Build a full login/signup/forgot-password flow with Supabase Auth
- Add Google Sign-In to an existing app
- Implement biometric login as a secondary auth method
- Set up realtime data sync for a chat-like feature

ASSIGNMENTS

- Build a complete authenticated app: signup, login, protected routes, profile editing, and logout, backed by Supabase or Firebase
- Write and connect a small custom Node.js API endpoint to the app for a feature of your choice

PROJECTS

- 'Social Habit Tracker' - authenticated app with email + Google login, user profiles stored in Supabase, and realtime updates when habits are marked complete

WEEKLY LEARNING OUTCOMES

- Production-ready authentication implementation
- Working social login integration
- Hands-on experience with a real backend-as-a-service
- Ability to connect to and consume any REST/GraphQL API

WEEKLY ASSESSMENT

Full auth flow demo including edge cases (expired token, wrong password, network failure)

Payments, Monetization & Third-Party Services

Integrate payment processing, in-app purchases, and common third-party SDKs used in production apps.

LEARNING OBJECTIVES

- Integrate a payment gateway for one-time and recurring payments
- Implement in-app purchases and subscriptions
- Add analytics and crash reporting
- Integrate maps, chat, or media SDKs as needed by app type

DETAILED TOPIC BREAKDOWN

8.1 Payment Gateway Integration

- Stripe React Native SDK setup
- Building a PaymentSheet checkout flow
- Handling payment success/failure/cancellation states
- Paystack/Flutterwave integration for African markets
- Webhooks and server-side payment verification (conceptual)
- PCI compliance basics and why you never store card data client-side

8.2 In-App Purchases & Subscriptions

- Apple In-App Purchase and Google Play Billing overview
- RevenueCat setup for cross-platform subscription management
- Configuring products and entitlements
- Restoring purchases and handling receipt validation
- Free trial and paywall UX patterns
- Testing IAP with sandbox accounts

8.3 Analytics & Monitoring

- Setting up Firebase Analytics or PostHog for mobile
- Tracking custom events (signup, purchase, screen_view)
- Sentry integration for crash and error reporting
- Performance monitoring basics
- Funnel analysis for onboarding and conversion
- Privacy-compliant analytics (App Tracking Transparency on iOS)

8.4 Common Third-Party SDKs

- Integrating a chat SDK (Stream Chat or similar) basics
- Embedding maps and geolocation-based search
- Social sharing with expo-sharing
- In-app reviews with expo-store-review
- Ad integration overview (AdMob) and monetization tradeoffs

HANDS-ON EXERCISES

- Implement a Stripe test-mode checkout for a mock product purchase
- Set up a paywall screen with RevenueCat sandbox subscription
- Add Sentry and trigger/capture a test crash
- Track 5 custom analytics events across an app flow

ASSIGNMENTS

- Add a complete monetization layer (one-time payment or subscription paywall) to a previous project app
- Set up crash reporting and analytics dashboards and submit a screenshot report of tracked events

PROJECTS

- 'Premium Fitness App' - free tier with limited workouts, paywall unlocking premium content via RevenueCat sandbox subscription, with analytics tracking and crash reporting wired in

WEEKLY LEARNING OUTCOMES

- Ability to integrate real payment and subscription systems
- Working analytics and crash reporting pipeline
- Understanding of app monetization strategies
- Experience wiring third-party SDKs into a React Native app

WEEKLY ASSESSMENT

Monetization flow demo + analytics dashboard review

Animations, Gestures & Advanced UI

Build fluid, native-feeling interactions using Reanimated and Gesture Handler.

LEARNING OBJECTIVES

- Build performant 60fps animations with Reanimated
- Implement gesture-driven interactions
- Create shared element and screen transitions
- Build advanced custom UI components

DETAILED TOPIC BREAKDOWN

9.1 React Native Reanimated Fundamentals

Shared values and useSharedValue useAnimatedStyle and worklets withTiming, withSpring, withDecay animation functions
Animating layout with entering/exiting animations interpolate() for value mapping Sequencing and chaining animations
Running animations on the UI thread vs JS thread

9.2 Gesture Handling

react-native-gesture-handler setup Pan, tap, pinch, and long-press gestures Combining gestures (simultaneous, exclusive, race)
Building swipe-to-delete list items Draggable and sortable list interactions Bottom sheet implementation with gestures
Pull-to-refresh custom gesture-based implementation

9.3 Screen & Shared Element Transitions

React Navigation screen transition customization Shared element transitions between screens
Custom modal presentation animations Parallax scroll effects Skeleton loading screens and shimmer effects
Lottie animations with lottie-react-native

9.4 Advanced Custom Components

Building a custom animated tab bar Building a custom animated bottom sheet from scratch
Building a swipeable card stack (Tinder-style) Building a custom pull-to-refresh animation
Accessibility considerations for animated components

HANDS-ON EXERCISES

- Build a swipe-to-delete list with spring animation
- Create a Tinder-style swipeable card stack
- Build a custom animated bottom sheet component
- Add Lottie success/error animations to a form submission flow

ASSIGNMENTS

- Build a fully animated onboarding flow with parallax effects and progress indicators
- Recreate one advanced UI pattern from a popular app (Instagram stories bar, Spotify now-playing bar, or similar) with full gesture support

PROJECTS

- 'Interactive Discovery App' - a Tinder-style swipe card app (jobs, recipes, or products) with animated card stack, gesture-based like/dislike, and animated transitions between screens

WEEKLY LEARNING OUTCOMES

- Fluency with Reanimated worklets and shared values
- Ability to build complex gesture-driven interactions
- Portfolio-quality animated UI components
- Understanding of animation performance on mobile

WEEKLY ASSESSMENT

Live gesture/animation demo graded on smoothness and code quality

Testing, Debugging & Performance Optimization

Ensure app quality through automated testing, systematic debugging, and performance profiling.

LEARNING OBJECTIVES

- Write unit and component tests for React Native code
- Perform end-to-end testing of critical user flows
- Profile and optimize app performance
- Identify and fix memory leaks and unnecessary re-renders

DETAILED TOPIC BREAKDOWN

10.1 Unit & Component Testing

- Jest configuration for React Native/Expo
- Writing unit tests for utility functions and hooks
- React Native Testing Library setup
- Testing components: rendering, props, and user events
- Mocking API calls and native modules in tests
- Snapshot testing tradeoffs
- Test coverage reports and thresholds

10.2 End-to-End Testing

- Maestro setup for mobile E2E testing
- Writing E2E flows: login, checkout, navigation
- Detox as an alternative E2E framework overview
- Running E2E tests on CI
- Testing on multiple device sizes and OS versions

10.3 Performance Profiling

- Using React DevTools Profiler in React Native
- Identifying unnecessary re-renders with why-did-you-render
- FlatList performance tuning (getItemLayout, windowSize, removeClippedSubviews)
- Image optimization and lazy loading with expo-image
- Reducing bundle size and analyzing bundle with source-map-explorer
- Hermes engine and its performance benefits
- Memory leak detection and fixing common leak patterns (listeners, timers)

10.4 Debugging & Crash Resolution

- Reading and interpreting native crash logs (iOS/Android)
- Using Sentry breadcrumbs to trace crash causes
- Debugging native module linking issues
- Handling app freezes and ANRs (Application Not Responding)
- Systematic bug triage and reproduction workflow

HANDS-ON EXERCISES

- Write 15 unit tests covering a utility library and a custom hook
- Write component tests for a form with validation logic
- Create a Maestro E2E flow testing full login-to-checkout journey
- Profile a slow list screen and apply at least 3 optimizations

ASSIGNMENTS

- Achieve 70%+ test coverage on a previous project's core logic and submit the coverage report
- Take a deliberately unoptimized app screen, profile it, and write a before/after performance report

WEEKLY LEARNING OUTCOMES

- Confident writing unit, component, and E2E tests
- Ability to profile and systematically improve app performance
- Skill in reading crash logs and resolving native issues
- A demonstrable before/after performance case study

WEEKLY ASSESSMENT

Test suite review + performance report grading

PROJECTS

- 'QA & Performance Audit' - take the E-Commerce Catalog project from Week 4, add a full unit/component test suite, one Maestro E2E flow, and a documented performance optimization pass

Native Modules, CI/CD & App Store Preparation

Extend apps with custom native code when needed, automate builds, and prepare all assets required for store submission.

LEARNING OBJECTIVES

- Understand when and how to write custom native modules
- Configure EAS Build for automated iOS/Android builds
- Set up CI/CD pipelines for mobile apps
- Prepare all app store assets, metadata, and compliance requirements

DETAILED TOPIC BREAKDOWN

11.1 Native Modules & Config Plugins

- When to eject/prebuild vs stay in Expo managed workflow
- Writing a simple native module (Swift/Kotlin) overview
- Using Expo Config Plugins to modify native projects
- Installing and linking third-party native libraries
- Handling native dependency conflicts
- expo prebuild and continuous native generation (CNG)

11.2 EAS Build & Automated Builds

- EAS CLI setup and eas.json configuration
- Build profiles: development, preview, production
- Configuring iOS credentials (certificates, provisioning profiles)
- Configuring Android credentials (keystore management)
- Triggering builds and monitoring build logs
- Internal distribution builds for beta testers
- EAS Update for over-the-air JS updates

11.3 CI/CD for Mobile

- GitHub Actions workflow for automated testing on PRs
- Automating EAS builds from CI on merge to main
- Automated version and build number bumping
- Fastlane overview for release automation
- Branch strategies for mobile release cycles (main/staging/release)
- Automated changelog generation

11.4 App Store Asset Preparation

- App icon and splash screen requirements per platform
- Screenshot generation for multiple device sizes
- Writing an effective app store listing (title, keywords, description)
- Privacy policy and terms of service requirements
- Apple App Privacy 'nutrition label' declarations
- Google Play Data Safety form requirements
- Age ratings and content guidelines compliance

HANDS-ON EXERCISES

- Configure eas.json with development, preview, and production profiles
- Trigger an EAS internal distribution build and install it on a real device
- Set up a GitHub Actions workflow that runs tests on every pull request
- Generate app store screenshots for 3 different device sizes

ASSIGNMENTS

- Fully configure EAS Build and produce a working preview build installable via QR code
- Prepare a complete app store listing package: icon, screenshots, description, privacy policy draft, and Data Safety form answers

PROJECTS

- 'Release Readiness Package' - take your capstone-in-progress app and produce a signed preview build plus complete store listing assets ready for submission

WEEKLY LEARNING OUTCOMES

- Ability to configure and trigger production mobile builds
- Working CI pipeline for a mobile codebase
- Complete, compliant app store asset package
- Understanding of native module integration boundaries

WEEKLY ASSESSMENT

Build pipeline review + store asset package audit

Capstone: Build & Publish a Cross-Platform App

Plan, build, polish, and submit a complete original mobile application to the Apple App Store and/or Google Play Store.

LEARNING OBJECTIVES

- Independently scope and plan a production mobile app
- Apply all prior skills to build a polished, tested, monetized app
- Submit a real app for review to Apple App Store and/or Google Play Store
- Present and defend technical and product decisions

DETAILED TOPIC BREAKDOWN

12.1 Capstone Planning & Scoping

- Choosing a viable app idea and defining core MVP scope
- User flow mapping and wireframing
- Technical architecture planning document
- Choosing backend (Supabase/Firebase/custom API)
- Feature prioritization: must-have vs nice-to-have
- Setting a realistic build timeline across the week

12.2 Full Build Sprint

- Implementing navigation, auth, and core screens
- Integrating chosen backend and data layer
- Adding at least one native device feature
- Implementing offline support where relevant
- Adding animations and polish pass
- Writing tests for critical flows
- Adding analytics and crash reporting

12.3 QA, Polish & Compliance

- Full regression testing across iOS and Android
- Accessibility pass (labels, contrast, touch targets)
- Performance and bundle size final check
- Preparing privacy policy and legal compliance
- Final app icon, splash screen, and branding polish
- Edge case handling (no network, empty states, errors)

12.4 Submission & Launch

- Creating App Store Connect and Google Play Console listings
- Submitting production build via EAS Submit
- Responding to reviewer feedback and rejections
- Setting up post-launch monitoring dashboards
- Planning a v1.1 roadmap based on projected user feedback
- Writing release notes and marketing copy

12.5 Presentation & Portfolio

- Recording a polished app demo video
- Writing a case study for the portfolio (problem, process, outcome)
- Preparing a technical walkthrough presentation
- Publishing project on GitHub with a professional README
- Presenting to peers and instructors for final review

HANDS-ON EXERCISES

- Draft and get sign-off on a one-page capstone scope document
- Build and demo a working alpha version by mid-week

- Run a full manual QA pass using a written test checklist
- Record a 2-minute demo video of the finished app

ASSIGNMENTS

- Submit the completed app build (EAS build link/APK/TestFlight link) along with source code repository
- Submit a written case study document covering architecture, challenges, and outcomes

PROJECTS

- Capstone: an original, fully-featured cross-platform mobile app including authentication, backend integration, at least one native device feature, offline support, animations, tests, and a real submission to the Apple App Store and/or Google Play Store

WEEKLY LEARNING OUTCOMES

- A published (or submission-ready) real-world mobile app
- A complete professional portfolio case study
- Demonstrated end-to-end mobile development competency
- Confidence presenting and defending technical work publicly

WEEKLY ASSESSMENT

Final capstone demo day presentation, code review, and store submission verification